

# Distributional Jump Bridges for Financial Disclosure Response Modeling

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## Abstract

Financial disclosure models often fuse event text with market history and predict every market-response target through one shared path. This paper proposes Distributional Jump Bridges (DJB), a neural event architecture that first estimates a no-event response distribution from pre-disclosure market state, then lets SEC filing text transport the event response through bounded shifts in target means and predictive log-variances. A return-conservative variant (RC-DJB) adds a hard return-mean firewall, providing a falsifiable test of whether disclosure text should directly move near-term abnormal-return point forecasts. On a leakage-controlled public-data study with 7,236 SEC 8-K events, 7,236 matched controls, and 2,463 held-out rows from 98 large U.S. firms, DJB obtains the lowest multi-target MSE, 0.0552 with 95% ticker-clustered interval [0.0396, 0.0773], after aligning firm and market return windows. Its abnormal-return Rank IC has the largest point estimate, 0.0330, although the ticker-clustered interval includes zero [-0.0042, 0.0710]. Paired bootstrap comparisons show DJB improves MSE over concat fusion by 0.0114 [0.0092, 0.0135] and over RC-DJB by 0.0010 [0.0005, 0.0015]. Leave-one-ticker-out Rank IC remains positive for every held-out ticker deletion, with minimum 0.0263. RC-DJB bridge interventions raise event-row MSE by 0.0183, 0.0037, and 0.0025 when the bridge is removed, text is zeroed, or text is shuffled. The results support a response-transport modeling principle: disclosure text is most reliable as a distributional bridge for event risk, liquidity, and uncertainty, while return-ranking claims should be reported with ticker-clustered and leave-one-ticker stability audits.

## 1 Introduction

Public disclosures are timed interventions in the market information set [Fama, 1970, Beaver, 1968, Patell and Wolfson, 1984]. A filing can change volatility, liquidity, and uncertainty even when its effect on cross-sectional abnormal-return ranking is weak or unstable [Bamber, 1986, Kothari, 2001]. Standard multimodal forecasters typically concatenate price history and text embeddings, then predict all targets with a shared head [Vaswani et al., 2017, Lim et al., 2021, Chen et al., 2025]. This design asks the same text representation to explain risk response, volume response, uncertainty, and return ranking. In financial data, that coupling is a liability: return labels are noisy, text is high dimensional, and small leakage, data snooping, or backtest overfitting can look like alpha [Campbell et al., 1997, Gu et al., 2020, White, 2000, Bailey et al., 2014, López de Prado, 2018].

This paper studies a stricter architectural question: should disclosure text enter an event model as an unconstrained predictor of every outcome, or as a distributional transport operator layered on top of ordinary market dynamics? Distributional Jump Bridges (DJB) first predict a no-event Gaussian response distribution from pre-event price, volume, and market-relative features. They then use disclosure text to generate bounded shifts in target means and predictive log-variances. The return-conservative variant (RC-DJB) adds the sharper restriction  $\Delta\mu_{\text{return}} = 0$ , so text may move volatility, volume, and uncertainty but cannot directly overwrite the abnormal-return point forecast.

The contributions are:

1. A distributional bridge architecture for event-driven financial response modeling.

2. A leakage-controlled SEC 8-K experiment with matched no-event controls and public post-filing market outcomes.
3. An evaluation protocol that aligns firm and market return windows and reports both ticker-clustered and leave-one-ticker stability audits for Rank IC.
4. Evidence that DJB improves response-regression error over fusion and bottleneck baselines, with RC-DJB intervention tests isolating what the return-mean firewall changes.

## 2 Related Work

Classical event studies estimate abnormal returns around information releases [Fama et al., 1969, Brown and Warner, 1985, MacKinlay, 1997]. Disclosure and earnings-announcement studies document price, volatility, and trading-volume responses after public information releases [Beaver, 1968, Patell and Wolfson, 1984, Bamber, 1986, Boehmer et al., 1991, Kothari and Warner, 2007]. DJB keeps this information boundary but replaces scalar event-window estimation with a neural distributional response model; RC-DJB is the return-conservative diagnostic variant. The abnormal-return construction follows the market-model and factor-model tradition [Sharpe, 1964, Fama and French, 1993, Carhart, 1997, Kothari and Warner, 2007]; this study uses a SPY-adjusted public-data variant rather than CRSP/factor residuals.

The architecture is related to recurrent, state-space, and dynamical-system methods, including LSTMs and gated recurrent units [Hochreiter and Schmidhuber, 1997, Cho et al., 2014], Koopman operators [Koopman, 1931], neural differential equations [Chen et al., 2018], structured state-space sequence models [Gu et al., 2022], and selective state-space models [Gu and Dao, 2023, Wang et al., 2025]. It also draws on probabilistic forecasting and time-series foundation model work [Dawid, 1984, Gneiting and Raftery, 2007, Salinas et al., 2020, Lim et al., 2021, Nie et al., 2023, Wu et al., 2023a, Ansari et al., 2024, Wu et al., 2024, Jin et al., 2023]. Marked point processes model event arrivals [Hawkes, 1971, Mei and Eisner, 2017]; here the disclosure time is observed and the task is post-event response prediction.

Financial text contains information relevant to prices and fundamentals [Tetlock, 2007, Tetlock et al., 2008, Kogan et al., 2009, Loughran and McDonald, 2011, Cohen et al., 2020]. Recent financial NLP systems and benchmarks study domain language models, multimodal financial forecasting, SEC filing reasoning, and leakage in financial agents [Jegadeesh and Wu, 2013, Araci, 2019, Wu et al., 2023b, Yang et al., 2023, Kong et al., 2025, Chen et al., 2025, Islam et al., 2023, Jiang et al., 2026, Li et al., 2025]. RC-DJB is also adjacent to causal representation and counterfactual outcome models [Rubin, 1974, Rosenbaum and Rubin, 1983, Melnychuk et al., 2022, Feng et al., 2024]. DJB differs by routing event text through an explicit residual distributional transport module; RC-DJB further imposes a target-specific restriction in which disclosure text may transport response distribution components while the abnormal-return mean remains no-event anchored.

## 3 Method

Throughout, DJB denotes the unconstrained distributional bridge, and RC-DJB denotes the return-conservative ablation. Both models share the same no-event dynamics and disclosure bridge. DJB allows text to shift all three target means; RC-DJB keeps the abnormal-return mean fixed and uses text only for volatility, volume, and log-variance transport.

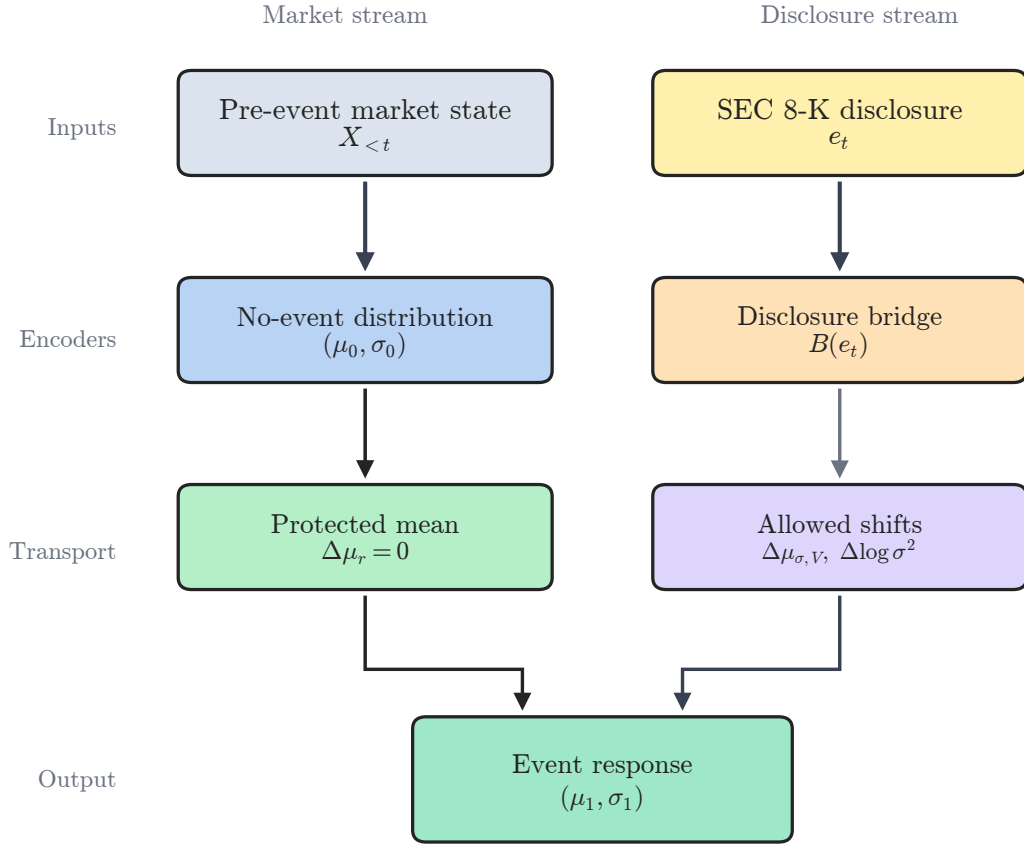


Figure 1: Distributional Jump Bridge architecture. The model first estimates a no-event response distribution from market history; disclosure text then generates bounded distributional shifts. The return-conservative diagnostic variant enforces  $\Delta\mu_r = 0$ .

**Inputs and targets.** Each sample  $i$  has label start date  $\tau_i$ , a 40-trading-day pre-event tensor  $X_i \in \mathbb{R}^{40 \times 8}$ , disclosure embedding  $e_i$ , metadata  $m_i \in \mathbb{R}^6$ , and target vector  $y_i \in \mathbb{R}^3$ . The targets are five-trading-day SPY-adjusted return, realized-volatility jump, and volume jump. Text-aware variants use chunk-pooled BGE-small disclosure embeddings, following the dense sentence and passage embedding literature [Reimers and Gurevych, 2019, Karpukhin et al., 2020, Xiao et al., 2023].

**Event-study labels.** Let  $P_{i,t}$ ,  $V_{i,t}$ , and  $r_{i,t}$  denote adjusted close, volume, and daily close return for firm  $i$ , and let  $r_t^M$  denote the SPY return. The label window  $\mathcal{W}_i = \{\tau_i, \dots, \eta_i\}$  starts on the disclosure date for before-close traded-day filings and on the next traded date otherwise;  $\eta_i$  is the fifth tradable session in the window. Let  $\mathcal{R}_i = \mathcal{W}_i \setminus \{\tau_i\}$  denote the close-to-close return dates strictly after the label-start close. The pre-window  $\mathcal{P}_i$  is the 20 traded sessions strictly before  $\tau_i$ . These choices follow the event-study convention that features must precede the public event time while outcomes are measured only after the event window begins [Brown and Warner, 1985, MacKinlay, 1997, Patell and Wolfson, 1984, Boehmer et al., 1991]. Realized-volatility differences are reported as a simple daily-data analogue of the realized-variation literature [Andersen et al., 2003]. The implemented targets are

$$y_{i,r} = \frac{P_{i,\eta_i}}{P_{i,\tau_i}} - 1 - \left( \prod_{t \in \mathcal{R}_i} (1 + r_t^M) - 1 \right), \quad (1)$$

$$y_{i,\sigma} = \text{sd}_{t \in \mathcal{W}_i}(r_{i,t}) - \text{sd}_{t \in \mathcal{P}_i}(r_{i,t}), \quad (2)$$

$$y_{i,V} = \frac{|\mathcal{P}_i|}{|\mathcal{W}_i|} \frac{\sum_{t \in \mathcal{W}_i} V_{i,t}}{\sum_{t \in \mathcal{P}_i} V_{i,t}} - 1. \quad (3)$$

The three targets are deliberately not identical event-study estimands. The return target is a conservative forward-drift label anchored at the label-start close; for a filing accepted before the market close, it therefore does not try to identify the intraday filing-day price jump from daily bars. A previous-close denominator would capture more immediate repricing, but with daily data it also mixes pre-filing intraday movement into the disclosure response. Volatility and volume are different response-distribution targets, so the filing-day session is included whenever the filing is public before the close. This convention tests whether disclosure text helps transport risk/liquidity response while leaving cross-sectional return ranking to the no-event state. Rows are admissible only when their maximum feature date is strictly before the label start,  $\max\{t : t \in X_i\} < \tau_i$ .

**No-event distribution.** A recurrent encoder maps pre-event market history to a latent state [Cho et al., 2014]:

$$h_i = \text{GRU}_\theta(X_i) + W_m m_i. \quad (4)$$

A bounded gate produces the no-event state  $\tilde{s}_i$ , from which the model predicts a diagonal Gaussian distribution, a standard proper-scoring-rule choice for probabilistic forecasts [Gneiting and Raftery, 2007, Salinas et al., 2020]:

$$\tilde{s}_i = h_i \odot (1 + 0.1 \tanh(g_\theta(h_i, m_i))), \quad (5)$$

$$[\mu_{0,i}, \ell_{0,i}] = f_\theta(\tilde{s}_i, m_i), \quad (6)$$

where  $\ell_{0,i}$  is log-variance.

**Distributional bridge.** Disclosure text generates a bounded latent state jump and distributional transport:

$$b_i = B_\theta(e_i), \quad (7)$$

$$j_i = \alpha \tanh(u_\theta(b_i)) \odot \sigma(v_\theta(b_i)), \quad (8)$$

$$[\Delta\mu_i, \Delta\ell_i] = T_\theta(\text{LN}(\tilde{s}_i + j_i), b_i, m_i). \quad (9)$$

Here LN denotes layer normalization [Ba et al., 2016]. The return-conservative architectural constraint is  $\Delta\mu_{i,\text{return}} = 0$ . For RC-DJB,

$$\mu_{1,i} = \mu_{0,i} + [0, \Delta\mu_{i,\text{volatility}}, \Delta\mu_{i,\text{volume}}], \quad \ell_{1,i} = \text{clip}(\ell_{0,i} + \Delta\ell_i). \quad (10)$$

DJB removes the zero-return-mean restriction and uses the full  $\mu_{1,i} = \mu_{0,i} + \Delta\mu_i$ . Both variants predict  $p_\theta(y_i) = \mathcal{N}(\mu_{1,i}, \text{diag}(\exp(\ell_{1,i})))$ .

**Objective.** For target dimension  $k$ , the Gaussian negative log likelihood used for probabilistic training and evaluation is a proper likelihood-based scoring objective [Gneiting and Raftery, 2007]:

$$\mathcal{L}_{NLL} = \frac{1}{2NK} \sum_{i=1}^N \sum_{k=1}^K [\exp(-\ell_{1,ik})(y_{ik} - \mu_{1,ik})^2 + \ell_{1,ik}], \quad K = 3, \quad \ell_{1,ik} \in [-7, 3]. \quad (11)$$

The pairwise rank regularizer is computed only over real-disclosure pairs in a minibatch, following pairwise learning-to-rank objectives [Burges et al., 2005, Cao et al., 2007]:

$$\mathcal{L}_{rank} = \frac{1}{|\mathcal{Q}|} \sum_{(i,j) \in \mathcal{Q}} \log \left( 1 + \exp \left[ -\text{sign}(y_{i,r} - y_{j,r}) \frac{\mu_{1,i,r} - \mu_{1,j,r}}{T} \right] \right), \quad (12)$$

where  $\mathcal{Q}$  contains event pairs with unequal abnormal-return labels and  $T = 0.02$ . The supervised term is the weighted Smooth-L1 loss after optional target standardization [Huber, 1964]:

$$\mathcal{L}_{Huber} = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K w_k \text{SmoothL1}_\beta \left( \frac{\mu_{1,ik} - \bar{y}_k}{s_k}, \frac{y_{ik} - \bar{y}_k}{s_k} \right). \quad (13)$$

For bottleneck models with posterior parameters  $(\mu_i^z, \ell_i^z)$ , the KL penalty follows variational and information-bottleneck regularization [Kingma and Welling, 2013, Tishby et al., 2000],

$$\mathcal{L}_{KL} = -\frac{1}{2} \text{mean}_{i,k} (1 + \ell_{ik}^z - (\mu_{ik}^z)^2 - \exp(\ell_{ik}^z)). \quad (14)$$

The control and consistency penalties are

$$\mathcal{L}_{control} = \frac{\sum_i \mathbb{I}\{a_i = 0\} \|\Delta \mu_i\|_2^2}{\sum_i \mathbb{I}\{a_i = 0\}}, \quad (15)$$

$$\mathcal{L}_{consistency} = \frac{1}{K} \sum_{k=1}^K \text{Var}_i(\Delta \mu_{i,k}), \quad (16)$$

where  $a_i = 1$  for real disclosures and  $a_i = 0$  for matched controls. The full training loss combines target-standardized Huber regression, Gaussian likelihood, control suppression, bottleneck regularization, bridge sparsity, residual consistency, and rank learning:

$$\mathcal{L} = \lambda_y \mathcal{L}_{Huber} + \lambda_n \mathcal{L}_{NLL} + \lambda_c \mathcal{L}_{control} + \lambda_{KL} \mathcal{L}_{KL} + \lambda_s \|z_i\|_1 + \lambda_v \mathcal{L}_{consistency} + \lambda_r \mathcal{L}_{rank}. \quad (17)$$

Here  $z_i$  is the stored disclosure-bridge latent used for sparsity diagnostics. The main run uses  $w_y = (3.0, 1.0, 0.25)$ ,  $\lambda_n = 0.10$ ,  $\lambda_c = 0.35$ ,  $\lambda_{KL} = 0.0$ ,  $\lambda_s = 0.001$ ,  $\lambda_v = 0.05$ , and  $\lambda_r = 0.15$ .

## 4 Data and Leakage Controls

The experiment uses public SEC EDGAR APIs [U.S. Securities and Exchange Commission, 2026]. The sample covers 98 usable large U.S. firms, SEC 8-K filings from 2020–2025, and public daily prices. Tickers are mapped to CIKs through the SEC company-tickers file. Filings are read from SEC submissions JSON, including archived submissions, using `acceptanceDateTime` and the primary EDGAR document URL. The parser stores source URL, accession number, accepted timestamp, available timestamp, extracted text, and text hash. Prior filing-text studies motivate using public SEC text as a market-response substrate [Loughran and McDonald, 2011, Jegadeesh and Wu, 2013, Cohen et al., 2020].

Daily prices are downloaded from public price endpoints, with SPY used as the market proxy for abnormal returns. The trading calendar is inferred from observed price rows. Event label windows start on the same observed trading day for filings accepted before 16:00 ET; filings accepted after 16:00 ET or on non-trading days start on the next observed trading day. Features use only prices strictly before the label start date. Labels use future returns only as targets. One deterministic same-ticker no-event control is constructed

for each filing, excluding dates within a 10-calendar-day blackout window around known events. The no-lookahead rule and blackout controls are motivated by standard event-study design and known financial backtesting leakage failure modes [MacKinlay, 1997, White, 2000, Hansen, 2005, Bailey et al., 2014, López de Prado, 2018, Li et al., 2025]. Figure 2 visualizes the chronological sample and the feature-to-label guard margin used by every row.

Table 1: Main public-data sample.

| Artifact                  | Count   |
|---------------------------|---------|
| Usable companies          | 98      |
| SEC 8-K events            | 7,236   |
| Matched no-event controls | 7,236   |
| Feature rows              | 14,472  |
| Public price rows         | 159,093 |
| Held-out rows             | 2,463   |

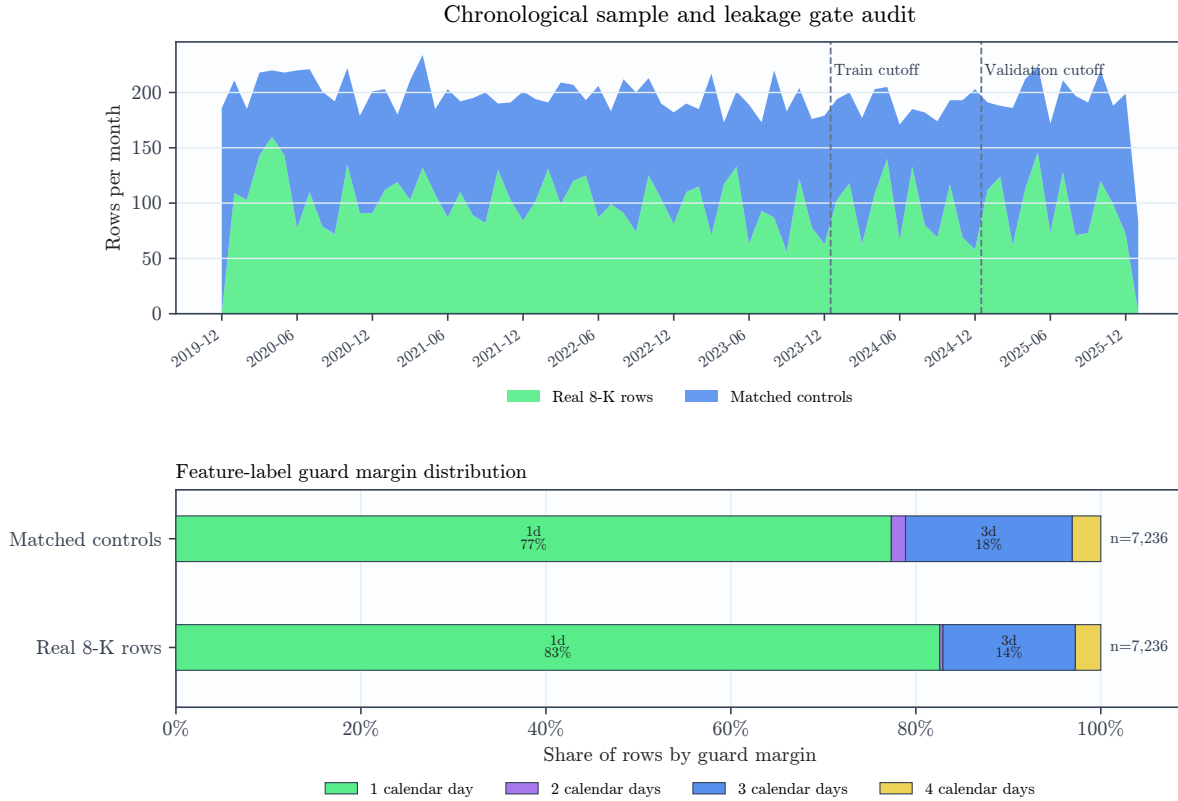


Figure 2: Chronological data substrate and feature-label leakage audit. The upper panel shows monthly real-event and matched-control rows with chronological split boundaries; the lower panel shows the share of rows by positive feature-label guard margin and the zero-violation leakage audit.

## 5 Experiments

Rows with  $\tau_i \leq 2023-12-31$  are training rows, rows from 2024 are validation rows, and later rows are held out. This yields 9,729 training rows, 2,280 validation rows, and 2,463 held-out rows across 98 held-out tickers and 13 held-out label-start months. Checkpoints are selected by the declared validation criterion for each run and evaluated once on the held-out split. All models use fixed seed 29. The chronological split follows time-series

evaluation practice, where future observations must not influence model selection or feature construction [Bergmeir and Benítez, 2012, Hyndman and Athanasopoulos, 2021, López de Prado, 2018].

Baselines include no-event dynamics, text-only MLP, concat fusion, Counterfactual Event Bottleneck Transformer (CEBT), Disclosure Operator Transformer (DOT), Event-Jump State Space Model (EJSSM), and an unconstrained Distributional Jump Bridge (DJB). The No-event baseline is trained on the same chronological event/control rows and event-window labels as the other models; it removes disclosure text rather than changing the training substrate. Confidence intervals use percentile bootstrap resampling; ticker-clustered intervals resample tickers with replacement [Efron, 1979, Efron and Tibshirani, 1993, Davison and Hinkley, 1997, Cameron et al., 2008, Cameron and Miller, 2015]. Paired comparisons use row-paired bootstrap resampling. Rank IC is Spearman correlation between predicted and realized abnormal return [Spearman, 1904, Grinold and Kahn, 2000].

**Evaluation statistics.** For model  $m$ , the reported multi-target error is

$$\text{MSE}(m) = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K \left( y_{ik} - \hat{y}_{ik}^{(m)} \right)^2. \quad (18)$$

Rank IC is the Spearman correlation between predicted and realized abnormal return:

$$\text{IC}(m) = \rho_P \left( \text{rank}(\hat{y}_{1:N,r}^{(m)}), \text{rank}(y_{1:N,r}) \right). \quad (19)$$

For a statistic  $S$ , bootstrap replicates  $S^{*(1)}, \dots, S^{*(B)}$  form percentile intervals

$$\text{CI}_{0.95}(S) = \left[ Q_{0.025} \{ S^{*(b)} \}_{b=1}^B, Q_{0.975} \{ S^{*(b)} \}_{b=1}^B \right], \quad (20)$$

The model-level intervals in Table 2 use  $B = 1000$ . Ticker-clustered intervals sample tickers with replacement and include all held-out rows for each sampled ticker. Rank IC uses integer ranks of predicted and realized abnormal returns. Paired MSE comparisons against DJB use  $B = 2000$  row-paired bootstrap replicates and

$$\Delta_{\text{MSE}}(m) = \frac{1}{N} \sum_{i=1}^N \left[ \frac{1}{K} \|y_i - \hat{y}_i^{(m)}\|_2^2 - \frac{1}{K} \|y_i - \hat{y}_i^{\text{DJB}}\|_2^2 \right], \quad (21)$$

with rows resampled as matched pairs. Paired rank comparisons use the same joined rows but report  $\text{IC}(\text{DJB}) - \text{IC}(m)$ .

**Probabilistic diagnostics.** For nominal level  $1 - \alpha$ , the diagonal Gaussian predictive interval for target  $k$  is

$$I_{ik}(1 - \alpha) = [\mu_{1,ik} - z_{1-\alpha/2} \exp(\ell_{1,ik}/2), \mu_{1,ik} + z_{1-\alpha/2} \exp(\ell_{1,ik}/2)], \quad (22)$$

and empirical coverage is

$$\hat{C}(1 - \alpha) = \frac{1}{NK} \sum_{i=1}^N \sum_{k=1}^K \mathbb{1}\{y_{ik} \in I_{ik}(1 - \alpha)\}. \quad (23)$$

Mean predictive standard deviation is  $\frac{1}{NK} \sum_{i,k} \exp(\ell_{1,ik}/2)$ . Interval coverage is reported because calibrated predictive uncertainty is distinct from point forecast error [Winkler, 1972, Gneiting and Raftery, 2007, Gneiting et al., 2007, Gneiting and Katzfuss, 2014].

Table 2: Held-out results after firm/market return-window alignment. MSE is over abnormal return, volatility jump, and volume jump. Intervals are 95% ticker-clustered bootstrap intervals.

| Model         | MSE           | MSE CI           | Rank IC       | Rank IC CI        |
|---------------|---------------|------------------|---------------|-------------------|
| No-event      | 0.0695        | [0.0517, 0.0933] | -0.0393       | [-0.0886, 0.0064] |
| Concat fusion | 0.0666        | [0.0499, 0.0886] | 0.0007        | [-0.0422, 0.0389] |
| CEBT          | 0.0696        | [0.0524, 0.0923] | -0.0295       | [-0.0752, 0.0160] |
| DOT           | 0.0698        | [0.0526, 0.0926] | -0.0043       | [-0.0487, 0.0422] |
| EJSSM-BGE     | 0.0566        | [0.0406, 0.0788] | 0.0254        | [-0.0107, 0.0633] |
| DJB           | <b>0.0552</b> | [0.0396, 0.0773] | <b>0.0330</b> | [-0.0042, 0.0710] |
| RC-DJB        | 0.0562        | [0.0404, 0.0783] | 0.0154        | [-0.0288, 0.0559] |

Table 3: Paired comparisons against DJB. Positive MSE values mean DJB has lower MSE. Positive rank values mean DJB has higher rank IC.

| Baseline      | MSE Diff. | 95% CI             | Rank IC Diff. | 95% CI            |
|---------------|-----------|--------------------|---------------|-------------------|
| No-event      | 0.01424   | [0.01145, 0.01765] | 0.0722        | [0.0204, 0.1293]  |
| Concat fusion | 0.01140   | [0.00919, 0.01353] | 0.0323        | [-0.0214, 0.0875] |
| CEBT          | 0.01443   | [0.01182, 0.01727] | 0.0625        | [0.0144, 0.1151]  |
| DOT           | 0.01459   | [0.01200, 0.01735] | 0.0372        | [-0.0257, 0.0993] |
| EJSSM-BGE     | 0.00136   | [0.00085, 0.00188] | 0.0076        | [-0.0192, 0.0370] |
| RC-DJB        | 0.00097   | [0.00052, 0.00146] | 0.0176        | [-0.0174, 0.0557] |

Table 4: Rank-temperature sensitivity. LOTO is the minimum leave-one-ticker-out Rank IC over the 98 held-out tickers.

| Model  | Temperature | MSE    | Rank IC | Rank IC CI        | LOTO min |
|--------|-------------|--------|---------|-------------------|----------|
| DJB    | $T = 0.02$  | 0.0552 | 0.0330  | [-0.0042, 0.0710] | 0.0263   |
| DJB    | $T = 0.10$  | 0.0559 | 0.0152  | [-0.0239, 0.0537] | 0.0086   |
| RC-DJB | $T = 0.02$  | 0.0562 | 0.0154  | [-0.0288, 0.0559] | 0.0095   |
| RC-DJB | $T = 0.10$  | 0.0562 | 0.0201  | [-0.0207, 0.0594] | 0.0146   |

**Bridge diagnostics.** The target-specific response-transport statistic for group  $c \in \{0, 1\}$  and target  $k$  is

$$T_{c,k} = \text{mean}_{i:a_i=c} |\Delta\mu_{i,k}|, \quad (24)$$

where  $c = 1$  denotes real 8-K rows and  $c = 0$  denotes controls. The aggregate transport reported below is  $\bar{T}_c = K^{-1} \sum_k T_{c,k}$ . Response-channel AUC uses  $\frac{1}{2}(|\Delta\mu_{i,\sigma}| + |\Delta\mu_{i,v}|)$  to classify real disclosures against matched controls. Latent bridge AUC uses the score  $s_i = \text{mean}_k |z_{i,k}|$ , with average ranks used for tied scores [Hanley and McNeil, 1982].

Table 5: Distributional bridge diagnostics on the held-out split. Log-variance Rank IC audits test whether return uncertainty shifts carry signed return-ranking information.

| Diagnostic                             | DJB     | RC-DJB  |
|----------------------------------------|---------|---------|
| Gaussian NLL                           | -2.2815 | -2.2961 |
| Observed 80% interval coverage         | 0.8948  | 0.9012  |
| Observed 95% interval coverage         | 0.9636  | 0.9632  |
| Mean predictive standard deviation     | 0.1351  | 0.1390  |
| Mean response transport, real 8-Ks     | 0.0235  | 0.0832  |
| Mean response transport, controls      | 0.0075  | 0.0684  |
| Return-mean transport, real 8-Ks       | 0.0053  | 0.0000  |
| Return-mean transport, controls        | 0.0050  | 0.0000  |
| Volatility transport, real 8-Ks        | 0.0160  | 0.1977  |
| Volatility transport, controls         | 0.0116  | 0.1932  |
| Volume transport, real 8-Ks            | 0.0493  | 0.0521  |
| Volume transport, controls             | 0.0060  | 0.0121  |
| Response-channel transport AUC         | 0.9269  | 0.8393  |
| Volatility-channel transport AUC       | 0.9948  | 0.9813  |
| Volume-channel transport AUC           | 0.8801  | 0.7860  |
| Latent bridge AUC, events vs. controls | 0.5297  | 0.6427  |
| Return log-var delta signed Rank IC    | -0.0046 | 0.0274  |

The per-channel audit clarifies the RC-DJB transport profile. Its return-mean transport is exactly zero by construction, while the response bridge remains active in volatility and volume. The volatility channel shows high transport for both real disclosures and controls, whereas the volume channel separates real events more clearly. The aggregate control transport therefore reflects a specific response-channel behavior, not a violation of the return-mean firewall.



**Intervention definitions.** The no-bridge intervention sets the bridge state jump, bridge hidden state, mean delta, and log-variance delta to zero, so  $\mu_1 = \mu_0$  and  $\ell_1 = \ell_0$ . The zero-text intervention replaces  $e_i$  with the zero vector before disclosure routing. The shuffled-text intervention permutes disclosure embeddings across held-out rows. For RC-DJB,  $\Delta\mu_{return} = 0$  in every variant, so abnormal-return point predictions and rank IC are invariant to these bridge interventions. These ablations test a learned model component rather than identify causal market effects [Rubin, 1974, Imbens and Rubin, 2015].

Table 6: RC-DJB intervention tests. Rank IC is unchanged by bridge interventions because the return-mean firewall fixes the abnormal-return point prediction to the no-event mean. The final column reports latent bridge AUC, not a direct text-discrimination score.

| Variant       | All MSE       | Event MSE     | Event NLL      | Rank IC | Latent AUC |
|---------------|---------------|---------------|----------------|---------|------------|
| Full RC-DJB   | <b>0.0562</b> | <b>0.0852</b> | <b>-2.2189</b> | 0.0154  | 0.6427     |
| No bridge     | 0.0713        | 0.1035        | 4.8117         | 0.0154  | 0.5000     |
| Zero text     | 0.0580        | 0.0889        | -2.2119        | 0.0154  | 0.0067     |
| Shuffled text | 0.0574        | 0.0877        | -2.2160        | 0.0154  | 0.3554     |

The no-bridge NLL degradation indicates that the base distribution alone is poorly calibrated for event rows; the bridge is therefore important for event-response uncertainty as well as point error. The zero-text latent AUC should be interpreted narrowly. With a constant zero text input, the latent score reflects bridge biases and metadata interactions rather than disclosure-content separation, which explains the inverted event/control ordering.

Table 7: Paired event-row MSE stress tests. Positive values mean full RC-DJB has lower event-row MSE than the intervention.

| Intervention  | Event MSE increase | 95% paired CI    |
|---------------|--------------------|------------------|
| No bridge     | 0.0183             | [0.0165, 0.0203] |
| Zero text     | 0.0037             | [0.0025, 0.0051] |
| Shuffled text | 0.0025             | [0.0014, 0.0036] |

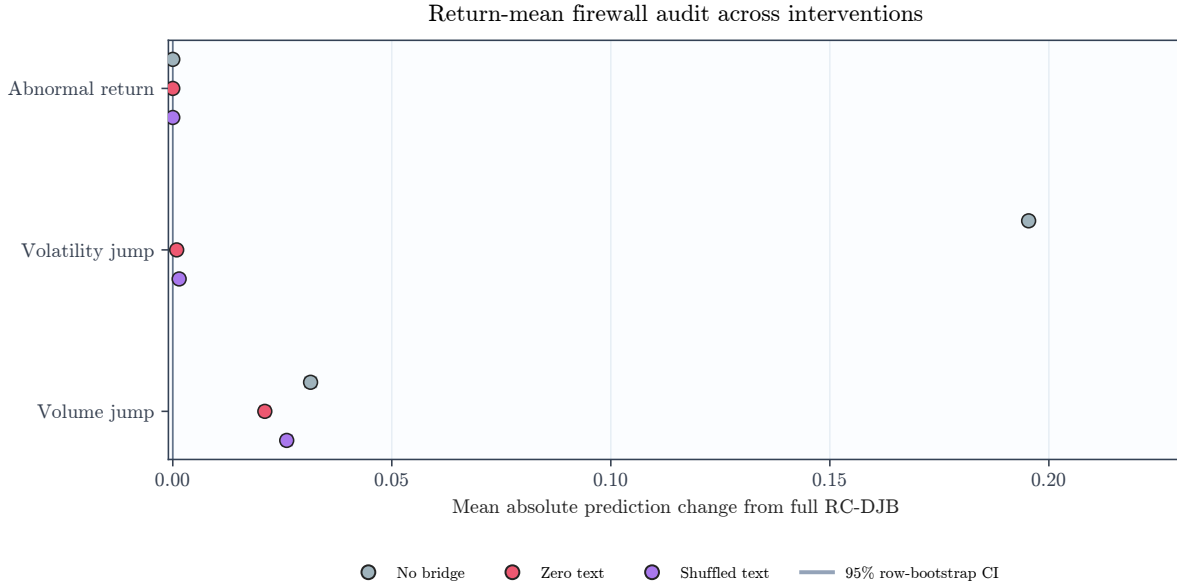


Figure 3: Return-mean firewall audit. Points show mean absolute point-prediction change from full RC-DJB under bridge interventions, with 95% row-bootstrap intervals. The abnormal-return mean prediction remains exactly invariant while response targets move; Table 5 separately audits return log-variance transport.

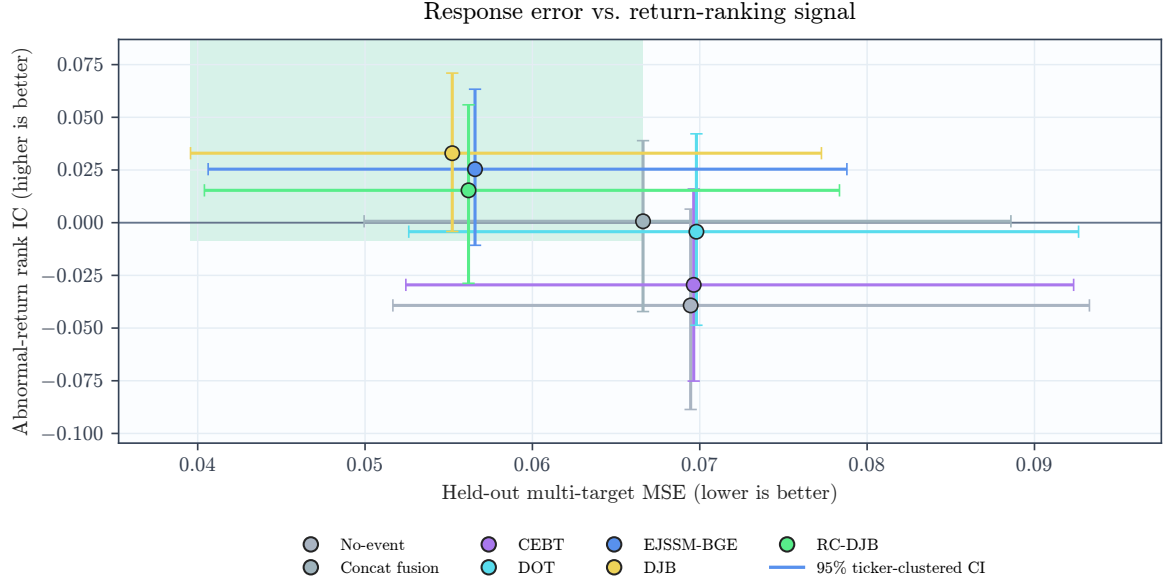


Figure 4: Response/ranking tradeoff after firm/market return-window alignment. DJB has the lowest held-out MSE and strongest point Rank IC, while ticker-clustered Rank IC intervals remain close to zero and require stability auditing.

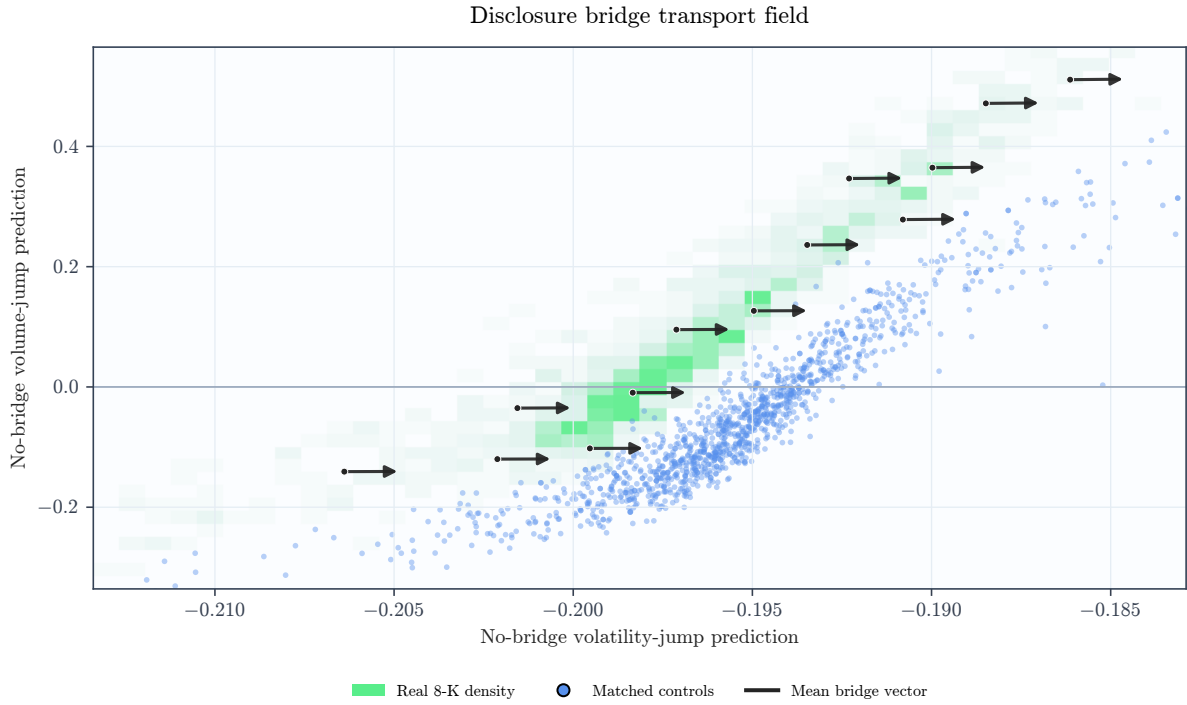


Figure 5: Held-out response-state transport induced by the disclosure bridge. The background shows real 8-K density in the no-bridge volatility/volume response plane, blue points show matched controls, and arrows show binned mean shifts in volatility-jump and volume-jump predictions.

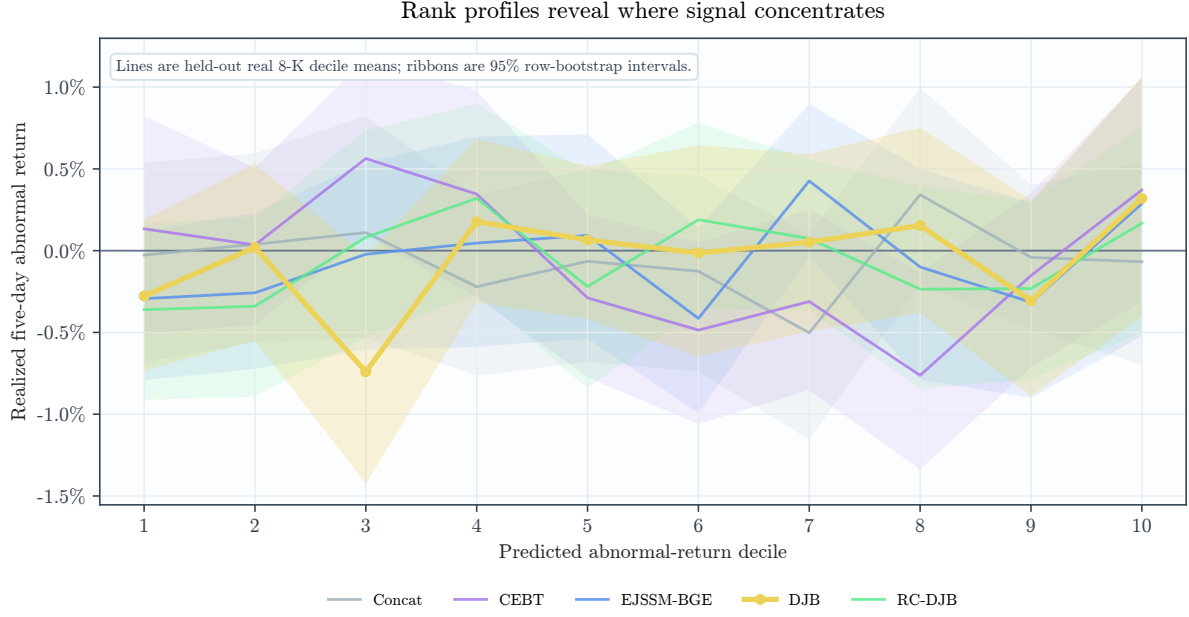


Figure 6: Predicted abnormal-return decile profiles on held-out real 8-K rows. Lines show realized five-day abnormal-return means; ribbons are 95% row-bootstrap confidence intervals computed within each predicted decile.

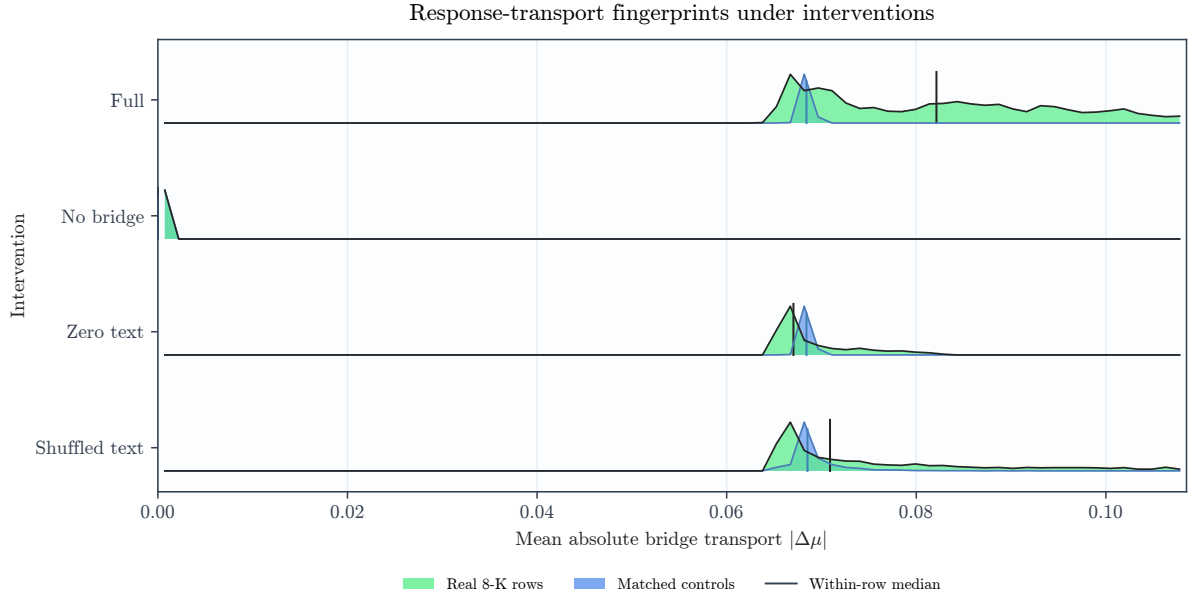


Figure 7: Response-transport fingerprints under bridge interventions. Ridge densities compare mean absolute response transport for real disclosure rows and matched controls; vertical marks show within-row medians.

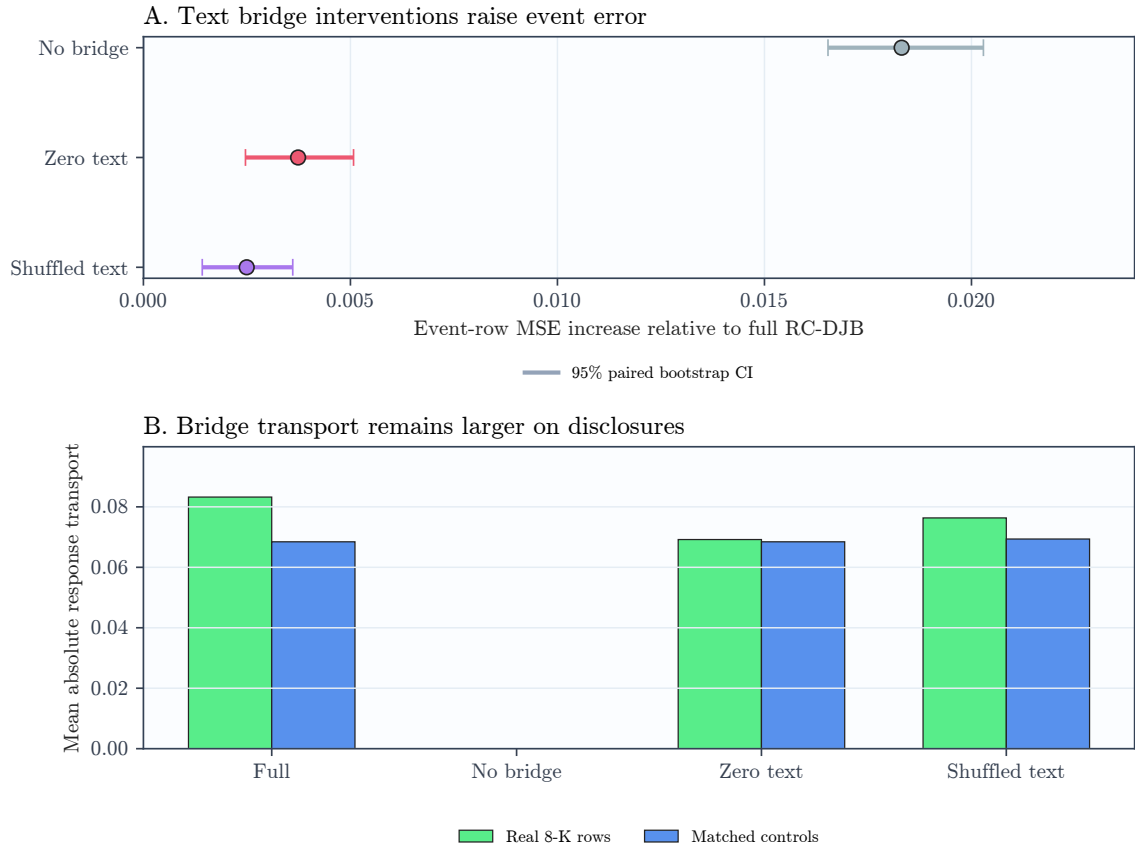


Figure 8: Interventions show that the disclosure bridge matters for event-response error. Rank IC does not move under text interventions because the return-mean firewall prevents direct text-to-return overwriting.

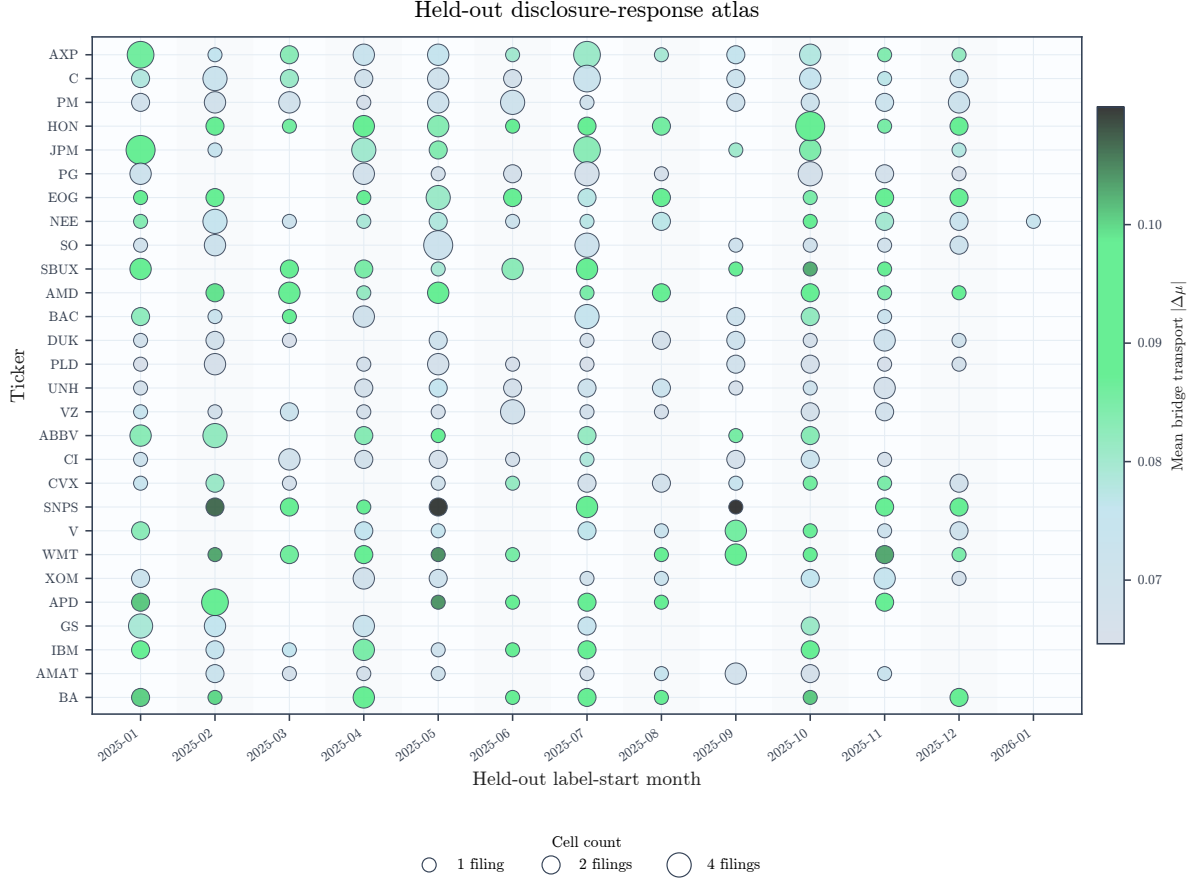


Figure 9: Held-out disclosure-response atlas for the most frequent test tickers. Marker area is the number of filings in the ticker-month cell and marker color is mean bridge transport, exposing heterogeneity across firms and calendar months.

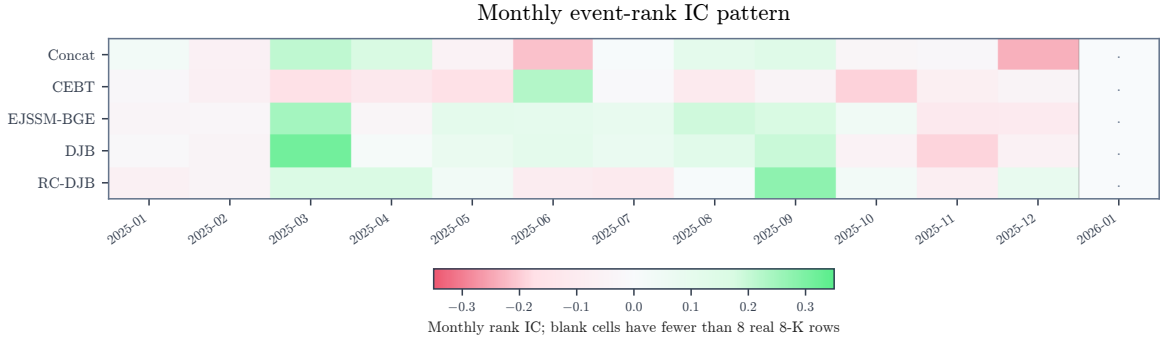


Figure 10: Monthly event-rank IC over held-out real 8-K rows. The model-level rank result is not uniform across months, motivating ticker-clustered intervals and leave-one-ticker-out stability checks rather than a single unqualified rank statistic.

## 6 Findings

**Distributional bridging is the main predictive gain.** After aligning the firm and market return windows, DJB obtains the best multi-target MSE (0.0552) and the strongest point Rank IC (0.0330). Paired

comparisons show lower MSE than every baseline, including concat fusion by 0.0114 [0.0092, 0.0135], EJSSM-BGE by 0.0014 [0.0009, 0.0019], and RC-DJB by 0.0010 [0.0005, 0.0015]. The gain is therefore not just a different training substrate; it is the learned residual distributional bridge.

**Return ranking is positive but statistically fragile.** DJB’s ticker-clustered Rank IC interval is [-0.0042, 0.0710], so the rank result should not be reported as cluster-significant. The leave-one-ticker-out audit is still informative: deleting any single held-out ticker leaves DJB Rank IC positive, with minimum 0.0263 after removing KLAC and maximum 0.0381 after removing COST. Figure 6 shows the same pattern visually: the rank profile contains signal, but it is not a monotone large-effect curve.

**The no-event baseline is not the DJB anchor.** The standalone No-event model has negative held-out rank IC, but this does not imply that pre-event market state is intrinsically anti-predictive. That baseline must explain disclosure-window returns, volatility, and volume using only ordinary market history and metadata. DJB trains its no-event state jointly with a disclosure bridge, control penalty, likelihood term, and rank objective. The empirical claim is therefore not that an isolated no-event forecaster is sufficient; it is that no-event dynamics become useful when event text is routed through residual response-distribution transport.

**The return-mean firewall is a diagnostic constraint.** RC-DJB is not the strongest predictive model after firm/market return-window alignment; it is a controlled ablation that asks what is lost when text is prevented from directly moving abnormal-return means. Its point Rank IC is lower than DJB’s (0.0154 vs. 0.0330), while its Gaussian NLL is slightly better (-2.2961 vs. -2.2815). This tradeoff clarifies the role of the firewall: it isolates risk, liquidity, and uncertainty transport from return-mean transport, but the unconstrained bridge is the stronger point forecaster in this sample.

**The RC-DJB bridge improves response modeling without moving return means.** Removing the RC-DJB bridge raises event-row MSE by 0.0183 [0.0165, 0.0203]; zeroing or shuffling text raises event-row MSE by 0.0037 [0.0025, 0.0051] and 0.0025 [0.0014, 0.0036]. These changes occur even though abnormal-return point predictions are unchanged by construction, which means the bridge is acting through volatility, volume, and uncertainty transport. The return log-variance delta has signed Rank IC 0.0274 in RC-DJB. The supported claim is therefore a return-mean firewall, not complete return-distribution isolation.

**Bridge diagnostics qualify the firewall interpretation.** The RC-DJB control penalty does not eliminate all response-channel movement on matched controls. Table 5 shows that return-mean transport is zero, but the volatility channel has similar magnitude on events and controls. This qualifies the interpretation of the control penalty: the firewall answers a target-specific routing question, while the per-channel transport audit shows where the bridge remains active.

**Rank-regularizer sensitivity is bounded.** The tighter rank temperature  $T = 0.02$  is not the sole source of the rank result. DJB with  $T = 0.10$  has lower Rank IC (0.0152), but RC-DJB moves from 0.0154 at  $T = 0.02$  to 0.0201 at  $T = 0.10$ . In both cases the ticker-clustered intervals cross zero. The appropriate conclusion is that the rank regularizer helps shape the ordering objective but does not create a cluster-significant trading claim.

**The bridge response is heterogeneous.** The ticker-month atlas in Figure 9 shows that bridge transport is not a uniform offset applied to all disclosures. Transport magnitude varies across firms and months, consistent with event-study evidence that disclosure response depends on the event, firm, and market state [Beaver, 1968, Patell and Wolfson, 1984, Kothari and Warner, 2007].

## 7 Reproducibility

The replication package includes the configurations, processed metadata, checkpoints, predictions, tables, and figures required to reproduce the reported results. Automated tests cover timestamp gating, after-hours

label starts, feature leakage rejection, control construction, deterministic tensor generation, finite model losses, probabilistic loss integration, and evaluation leakage guards. Labels are computed only from future price windows and are never included in model inputs.

## 8 Limitations

The study is intentionally scoped to a leakage-controlled architecture test. The 98-firm universe is a dense large-cap disclosure-response panel, not an all-equity trading universe; scaling the same protocol to broader S&P 500, Russell 3000, small-cap, and low-liquidity samples is the natural external-validity test. The price source is public daily data rather than CRSP or intraday trades and quotes, so the return label is designed as a conservative forward-drift target rather than an immediate jump estimator. A previous-close or intraday label would answer a complementary question about first-session repricing. Abnormal returns are SPY-adjusted rather than factor-model residuals [Fama and French, 1993, Carhart, 1997]. BGE-small is a general-purpose encoder with chunk mean pooling; domain-specific or long-document financial encoders such as FinBERT, BloombergGPT-style models, FinGPT, or SEC-specialized long-context encoders could change absolute response error [Araci, 2019, Wu et al., 2023b, Yang et al., 2023]. Because all text-aware models in the main experiment share the same frozen text representation, the reported comparison isolates the routing architecture rather than encoder scale. Rank IC is positive in point estimate and stable under leave-one-ticker deletion, but its ticker-clustered interval crosses zero; net performance after spreads, commissions, market impact, borrow constraints, capacity, and turnover requires a separate portfolio-construction study [Grinold and Kahn, 2000, Novy-Marx and Velikov, 2016, Frazzini et al., 2018, López de Prado, 2018]. The return-mean firewall is an architectural and predictive constraint, not structural causal identification [Rubin, 1974, Imbens and Rubin, 2015]; it blocks direct movement of the return mean but still allows return log-variance transport.

## 9 Ethics and Use

This paper studies event representations using public filings and public daily prices. The results do not constitute trading recommendations. The replication package is intended to support leakage auditing, independent reproduction, and falsification of the empirical claims.

## 10 Conclusion

DJB shows that the route by which disclosure text enters a financial event model matters. Disclosure text is most useful when it acts as a residual distributional transport operator layered on top of no-event market dynamics. In this public-data experiment, the unconstrained DJB is the strongest response forecaster, while RC-DJB clarifies what changes when a return-mean firewall is imposed. The resulting principle is architectural and testable: model disclosures as distributional jumps, audit return-ranking claims under ticker-clustered uncertainty, and separate return-mean transport from risk, liquidity, and uncertainty transport.

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